MTH 207 — §4.5 Curve Sketching Name _____

Example 1. Carried over from yesterday

Sketch $f(x) = x\sqrt{4-x^2}$ on $[-2, 2]$.

You are given (already simplified — do *not* spend time re-deriving these):

$$f'(x) = \frac{2\left(2-x^2\right)}{\sqrt{4-x^2}} \qquad f''(x) = \frac{2x\left(x^2-6\right)}{\left(4-x^2\right)^{3/2}}$$

1. Domain _____

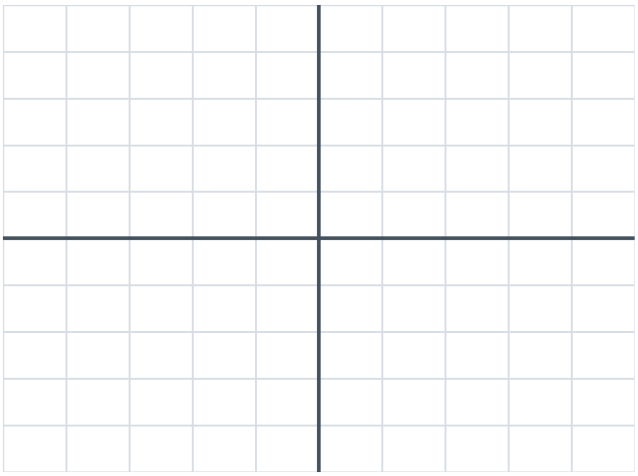
2. Intercepts _____

3. Symmetry _____

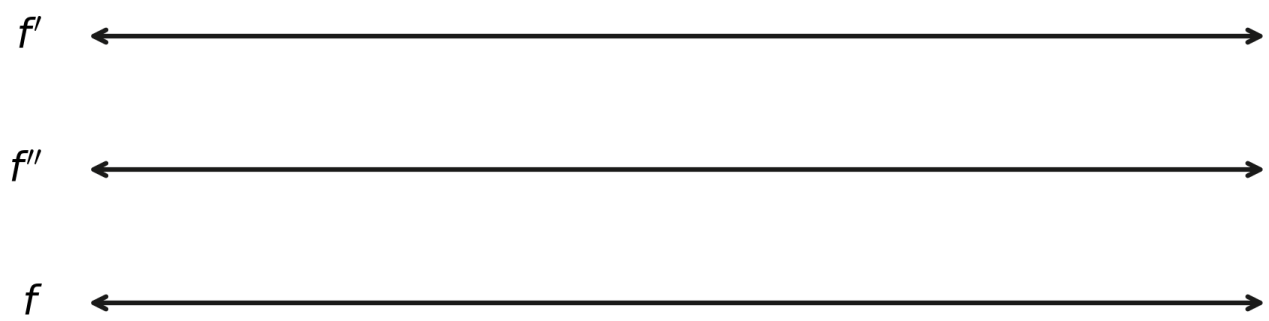
4. Asymptotes _____

6. Local extrema _____

7. Inflection points _____



5 and 7. Mark every cut from *either* chart, then one shape per piece.



Step 1 is real work here. And look at what f' does at the ends.

Work

Example 2. An asymptote on one side only

Sketch $f(x) = x^2 e^{-x}$.

You are given (already simplified — do *not* spend time re-deriving these):
$f'(x) = x(2-x)e^{-x}$ $\qquad$ $f''(x) = \left(x^2 - 4x + 2\right) e^{-x}$

1. Domain _____

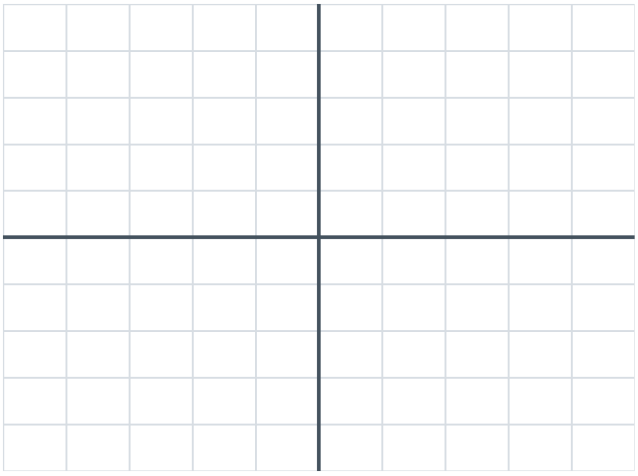
2. Intercepts _____

3. Symmetry _____

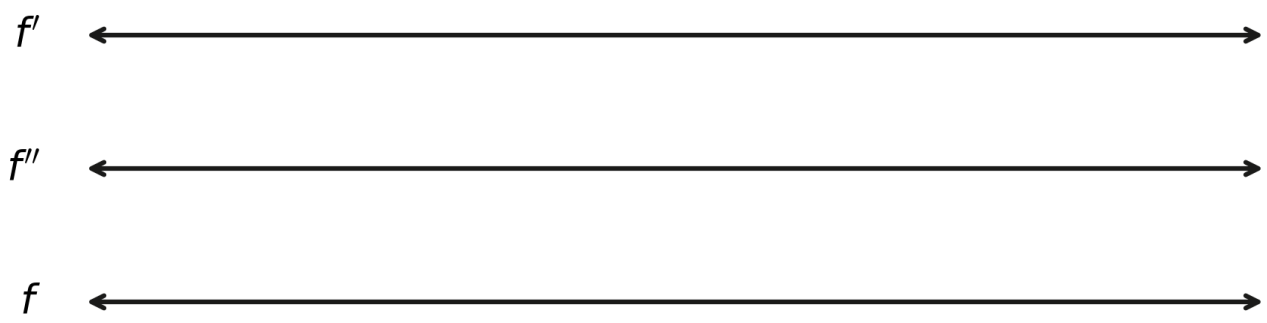
4. Asymptotes _____

6. Local extrema _____

7. Inflection points _____



5 and 7. Mark every cut from *either* chart, then one shape per piece.



e^{-x} is always positive, so it never decides a sign.

Work

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Example 3. Exactly what Exam 3 asks

Sketch $f(x) = \dfrac{x^2-4}{x^2+4}$.

You are given (already simplified — do *not* spend time re-deriving these):

$$f'(x) = \frac{16x}{(x^2+4)^2} \qquad f''(x) = \frac{-16\left(3x^2-4\right)}{(x^2+4)^3}$$

1. Domain _____

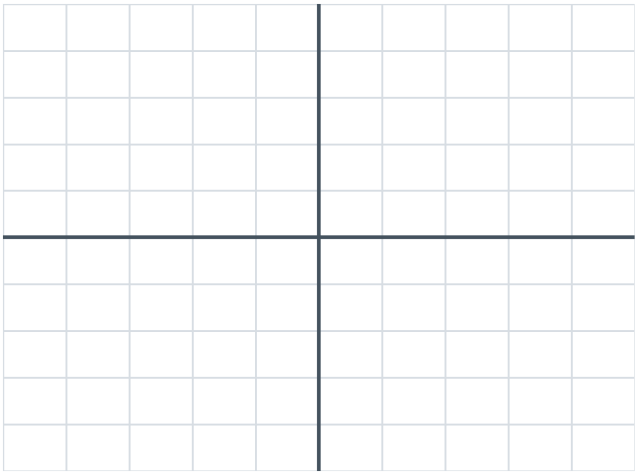
2. Intercepts _____

3. Symmetry _____

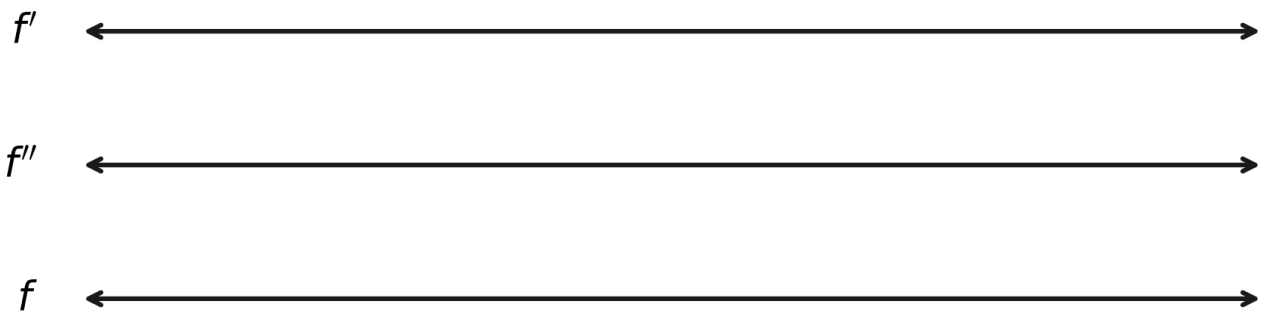
4. Asymptotes _____

6. Local extrema _____

7. Inflection points _____



5 and 7. Mark every cut from *either* chart, then one shape per piece.



Step 3 pays off here — do half the line and mirror it.

Work

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Example 4. A fractional exponent, and a cusp

Sketch $f(x) = x^{\frac{2}{3}}(x-5)$.

You are given (already simplified — do *not* spend time re-deriving these):

$f'(x) = \dfrac{5(x-2)}{3x^{1/3}}$ $\qquad$ $f''(x) = \dfrac{10(x+1)}{9x^{4/3}}$

1. Domain _____

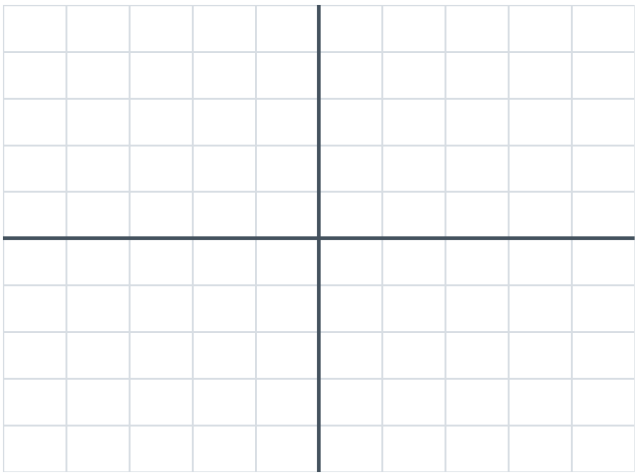
2. Intercepts _____

3. Symmetry _____

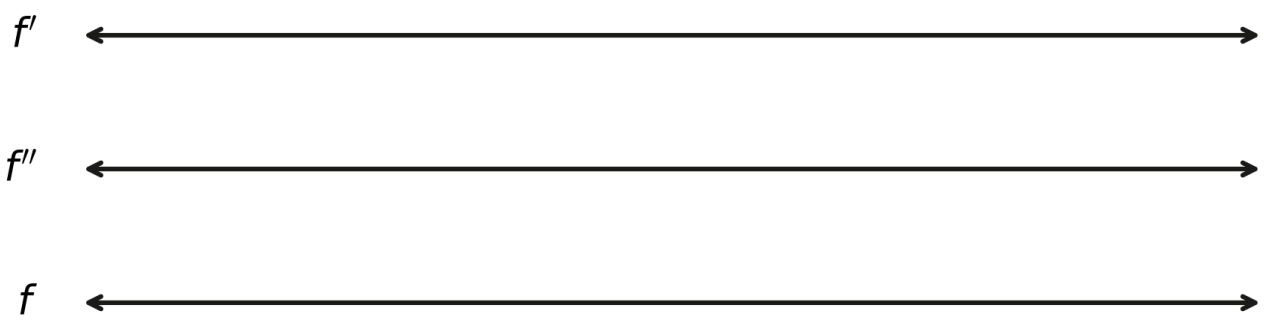
4. Asymptotes _____

6. Local extrema _____

7. Inflection points _____



5 and 7. Mark every cut from *either* chart, then one shape per piece.



Where is f' undefined? Is f still defined there?

Work

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Example 5. A trigonometric one

Sketch $f(x) = x + 2\cos(x)$ on $[0, 2\pi]$.

You are given (already simplified — do *not* spend time re-deriving these):
$f'(x) = 1 - 2\sin x$ $\qquad$ $f''(x) = -2\cos x$

1. Domain _____

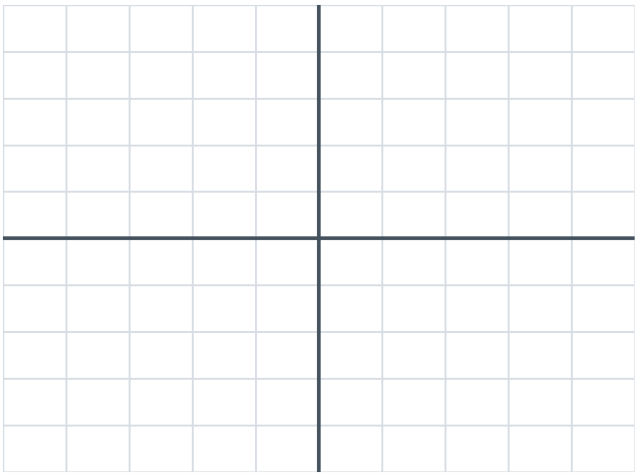
2. Intercepts _____

3. Symmetry _____

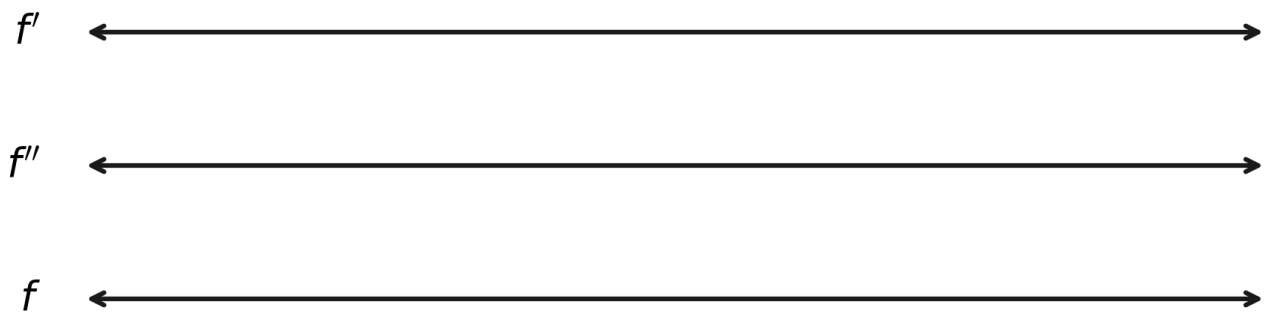
4. Asymptotes _____

6. Local extrema _____

7. Inflection points _____



5 and 7. Mark every cut from *either* chart, then one shape per piece.

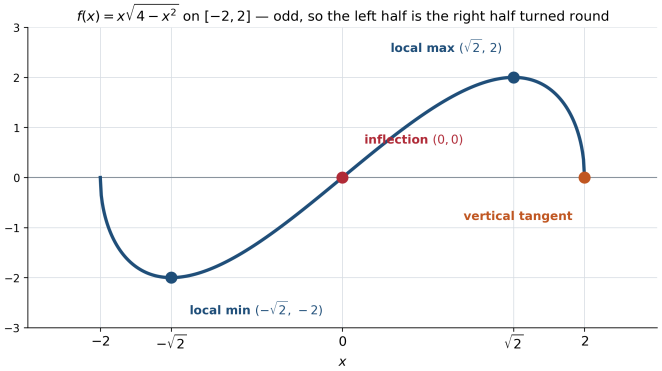


The critical numbers come from SOLVING, not factoring. Closed interval: the ends count.

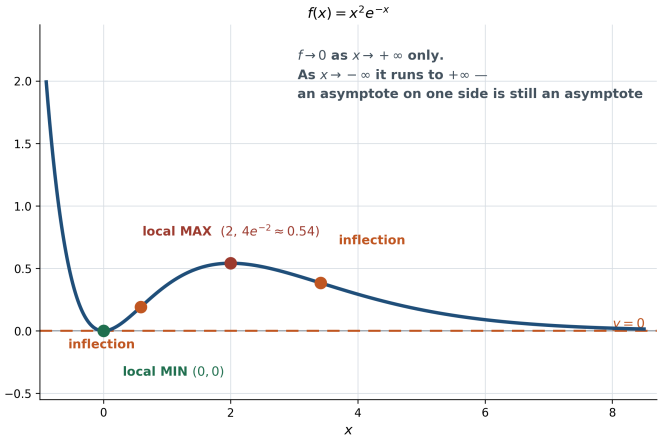
Work

Check your sketches against these. **Full worked solutions — all seven steps and both sign charts for every one — are at the link below.**

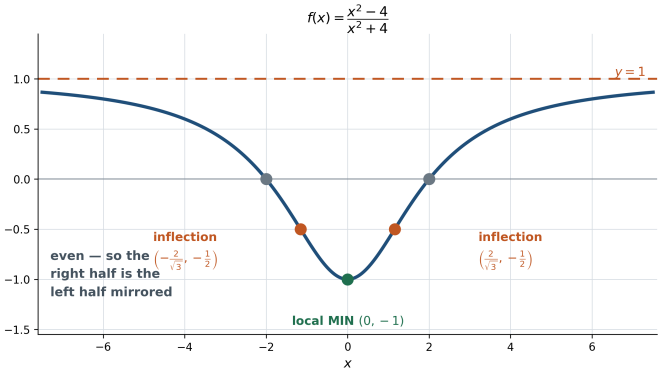
1. $x\sqrt{4-x^2}$



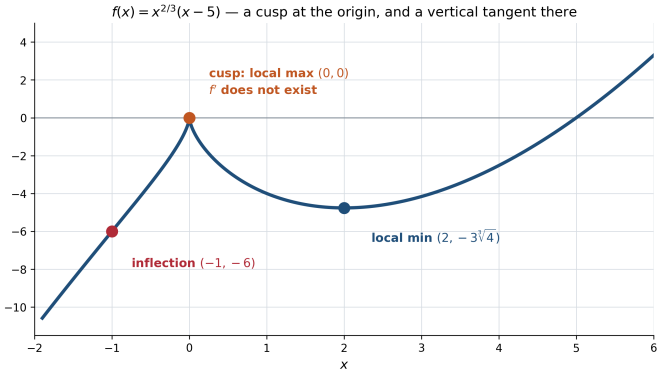
2. x^2e^{-x}



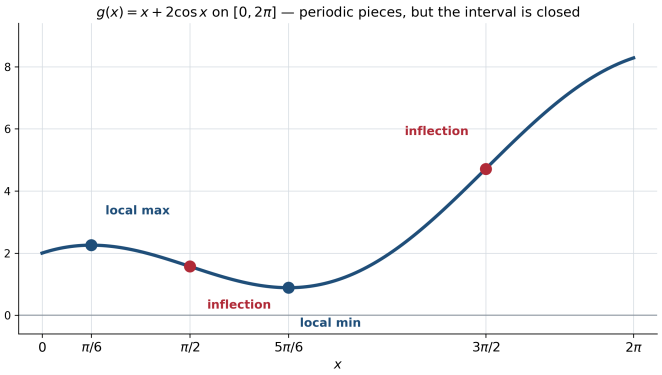
3. $\frac{x^2-4}{x^2+4}$



4. $x^{\frac{2}{3}}(x-5)$



5. $x + 2\cos(x)$



Worked solutions. Scan, or go to

`https://datascienceuwl.github.io/MTH207/handouts/curve-sketching-solutions.pdf`

Every example above, worked through all seven steps, with the f' and f'' sign charts filled in and the shape line assembled.